Cyber Security Task 02: CIA Triad & Risk Assessment

Objective The purpose of this task is to understand the core principles of Cyber Security and learn how organizations identify and manage security risks.

Part A: Understanding the CIA Triad

1. Confidentiality

Definition

Confidentiality means protecting information from being accessed by unauthorized people. Only authorized users should be able to view sensitive data.

Why is it Important?

Confidentiality helps protect personal, financial, and organizational information from theft or misuse. It ensures privacy and prevents confidential data from falling into the wrong hands.

Real-World Example

Online banking systems require usernames, passwords, and OTPs so that only the account owner can access their banking information.

2. Integrity

Definition

Integrity means ensuring that data remains accurate, complete, and unchanged unless modified by authorized users.

Why is it Important?

Integrity helps maintain trust in information. If data is altered accidentally or maliciously, it can lead to incorrect decisions, financial losses, or system failures.

Real-World Example

A university's examination system should store students' marks accurately. Unauthorized changes to marks would violate data integrity.

3. Availability

Definition

Availability means ensuring that systems, applications, and data are accessible whenever authorized users need them.

Why is it Important?

Availability ensures that users can access services without unnecessary interruptions. Downtime can affect productivity, business operations, and customer satisfaction.

Real-World Example

An online shopping website should remain accessible during sales events so that customers can browse and purchase products without issues.

Security Principle	Definition	Importance	Real-World Example
Confidentiality	Protects information from unauthorized access.	Maintains privacy and prevents data theft.	Online banking accounts protected by passwords and OTPs.
Integrity	Ensures data remains accurate and unaltered.	Maintains trust and prevents incorrect information.	Student marks stored securely in a university database.
Availability	Ensures systems and data are accessible when needed.	Keeps services running and minimizes downtime.	E-commerce websites remaining available during peak sales.

Part B: Real-World Case Study:

Real-World Case Study: WannaCry Ransomware Attack

1. What happened?

WannaCry was a global ransomware attack that occurred in May 2017. The malware spread rapidly by exploiting a vulnerability in Microsoft Windows systems. Once a computer was infected, the ransomware encrypted the user's files and displayed a message demanding payment in Bitcoin to restore access to the files. The attack affected more than 200,000 computers in over 150 countries, including hospitals, businesses, and government organizations.

2. Which part of the CIA Triad was affected?

The attack mainly affected **Availability** because users were unable to access their files and systems after the ransomware encrypted them. It also affected **Integrity** because important data was modified and made unusable by unauthorized software.

3. What was the impact?

The attack caused major disruptions worldwide. Hospitals had to cancel appointments and delay treatments because their computer systems became unavailable. Many businesses experienced downtime, financial losses, and operational interruptions. Organizations also spent significant amounts of money recovering systems and improving security measures.

4. How could it have been prevented?

The attack could have been prevented by regularly updating operating systems and installing security patches released by

Microsoft. Organizations should also maintain regular data backups, use antivirus software, and educate employees about cyber threats. Network monitoring and proper security practices could have reduced the impact of the attack and prevented the malware from spreading rapidly.

Part C: Risk Assessment Activity

Risk Assessment Table

Asset	Threat	Risk Level	Impact
Student Records	Data Breach	High	High
Examination Data	Unauthorized Access or Data Tampering	Medium	High
College Website	DDoS Attack or Website Defacement	Medium	High
Faculty Information	Phishing or Identity Theft	Medium	Medium
Network Infrastructure	Malware or Ransomware Attack	High	High

1. Student Records

Student records contain personal and academic information. A data breach could expose sensitive details, leading to privacy and security issues.

2. Examination Data

Examination data includes question papers, marks, and results. Unauthorized access or modification could affect the integrity of the examination process.

3. College Website

The college website provides information to students and staff. A cyberattack could make the website unavailable or display false information.

4. Faculty Information

Faculty details such as contact information and credentials should be protected. Phishing attacks can compromise accounts and lead to identity theft.

5. Network Infrastructure

The college network connects computers, servers, and internet services. Malware or ransomware attacks can disrupt operations and make resources inaccessible.